

FAMILY GROUP

ARSL SHANI &
AINA MALIK



Mcs Family Group 🎉

Created by Arsl, 3/30/20

@nameshadowart

CS614 GRAND QUIZ

SOLUTION MADE BY:

ARSL SHANI, AINA MALIK & MCS FAMILY GROUP TEAM

1) Online Extraction is a kind of ----- data extraction:

- Physical page 132
- Logical
- Dimensional
- Multi valued

- Physical Extraction
 - Online Extraction

2) The ----- saw the advent of disk storage, or DASD (direct Access Storage Device) :

- 1960s
- 1970s page 13
- 1950s
- 1990s

The 1970s saw the advent of disk storage, or DASD (Direct Access Storage Device).

3) In context of data warehouse, normally it becomes difficult to extract data from different sources because these sources are normally.

- Heterogeneous page 140
- Homogeneous
- Centralized
- Baseline

It is very difficult to extract the data from different source systems, because by definition they are heterogeneous, and as a consequence, there are multiple representations of the

4) Which of the following is not a task of Data Transformation?

- Conversion
- Summarization
- Enrichment
- Full Data Refresh page 135

16.6 Data Transformation

- Basic tasks
 - Selection
 - Splitting/Joining
 - Conversion
 - Summarization
 - Enrichment

5) Which of the following is not an Orr's Law of Data Quality?

- "Data that is not used cannot be corrected!"
- "Data quality is a function of its use, not its collection!"
- "Data will be no better than its most stringent use!"
- "Data duplication can be harmful for the organization!" page 181

21.3 Orr's Laws of Data Quality

Law #1: "Data that is not used cannot be correct!"

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Data Warehousing (CS614)

Law #2: "Data quality is a function of its use, not its collection!"

Law #3: "Data will be no better than its most stringent use!"

Law #4: "Data quality problems increase with the age of the system!"

Law #5: "The less likely something is to occur, the more traumatic it will be when it happens!"

6) Flat files are one of the prevalent structures used in ----- data extraction:

- Online
- Offline page 134
- Incremental
- Full

- **Offline Extraction**

- Data NOT extracted directly from the source system, instead staged explicitly outside the original source system.
- Data is either already structured or was created by an extraction routine.
- Some of the prevalent structures are:
 - Flat files

7) Which of the following is NOT one of the advantages of changed data capture (CDC) technique?

- Flat files are not required
- Limited query interface is required for data extraction page 152
- No incremental on-line I/O required for log tape
- Extraction of changed data occurs immediately

Some of the advantages of CDC are;

1. Database extraction from INSERT, UPDATE, and DELETE operations occurs immediately, at the same time the changes occur to the source tables.
2. In the absence of CDC database extraction is marginal at best for INSERT operations, and problematic for UPDATE and DELETE operations, because the data is no longer in the table.
3. Stages data directly to relational tables; there is no need to use flat files. In the absence of CDC, the entire contents of tables are moved into flat files.
4. Log tape processing requires no extra I/O in the middle of online processing, as does

8) The most common range partitioning is on

- Color
- Date page 66
- PhoneNo
- Name

The most common use of range partitioning is on date. This is especially true in data

9) A relation is said to be in first normal form(1NF), if it does not contain _____

- Single value column
- Multi-valued column page 43
- Derived column
- Composite column

Although the table is shown in 1NF i.e. it contains atomic values, there are no repeating values, there is no aggregation, yet it contains redundant data. For example, information

A table is in the first normal form (1NF) if

- The table has a primary key.
- No single column has multiple values.
- The nonprimary key columns depend on the primary key.

10) In a fully normalized database, too many _____ are required

- Values
- Joins page 49
- Queries
- Conditions

There should be a balance between normalized and de-normalized forms. In a fully normalized form, too many joins are required and in a totally de-normalized form, we

11) In the data warehouse, data is collection from ----- sources:

- Homogeneous
- Heterogeneous page 21
- External
- Internal

into a box and soon you have a Data Warehouse up and running. A Data Warehouse evolves over time, you don't buy it. Basically it is about taking/collecting data from different heterogeneous sources. Heterogeneous means not only the operating system is

12) De-normalization is more like a "controlled crash" with the aim to ----- without loss of information:

- Check
- Balance
- Decrease
- Enhance page 49

It is not chaos, more like a "controlled crash" with the aim of performance enhancement without loss of information.

13) ----- is making all efforts to increase effectiveness and efficiency in meeting and accepted customer expectation:

- Quality assurance
- Quality improvement page 183
- Quality maintenance
- Quality Establishment

Quality improvement is making all efforts to increase effectiveness and efficiency in meeting accepted customer expectations.

14) ----- is the application of intelligence and experience to get common goals.

- Wisdom page 11
- Education
- Power
- Information

Intelligence is appreciation of "why", and finally wisdom (not shown in the figure-1.1) is the application of intelligence and experience toward the attainment of common goals, and wise people are powerful. Remember knowledge is power.

15) In the data transformation, ----- is the rearrangement and simplification of individual

- Aggregation
- Enrichment page 136
- Splitting joining
- Conversion

Enrichment: This task is the rearrangement and simplification of individual fields to make them more useful for the data warehouse environment. You may use one or more

16) Grain of a fact table means :

- The meaning of one fact table row page 109
- The meaning of one dimensional table row
- Summary of aggregates in all fact tables
- Summary of aggregates in all dimension tables

Step-2: Grain of a fact table = the meaning of one fact table row. Determines the maximum level of detail of the warehouse.

17) Normalization ----- :

- Reduces redundancy page 41
- Increases redundancy
- Reduces joins
- Reduces tables

Normalization is the process of efficiently organizing data in a database by decomposing (splitting) a relational table into smaller tables by projection. There are basically two goals of normalization as follows:

1. Eliminate redundant data (for example, storing the same data in more than one table)

18) Which of the following is not an example of a typical grain :

- Individual transaction
- Daily aggregates
- Monthly aggregates
- Normalized attributes page 111

- Typical grains
 - Individual Transactions
 - Daily aggregates (snapshots)
 - Monthly aggregates

19) Multi-dimensional databases(MDDs) typically use ----- formats to store pre-summarized cube structures:

- SQL
- Proprietary file page 79
- Object oriented
- Non-proprietary file

Multi-dimensional databases (MDDs) typically use proprietary file formats to store pre-summarized cube structures. There are dozens of vendors who provide MDD products.

20) ----- provides a combination of “relational databases access” and “cube” data structures within a single framework:

- HOLAP page 78
- DOLAP
- MOLAP
- ROLAP

HOLAP provides a combination of relational database access and “cube” data structures within a single framework. The goal is to get the best of both MOLAP and ROLAP:

21) Data Warehouse provides the best support for analysis while OLAP carries out the ----- task:

- Mandatory
- Whole
- Analysis page 69
- Prediction

Data Warehouse provides the best support for analysis while OLAP carries out the analysis task. Although there is more to OLAP technology than data warehousing, the

22) ----- involves splitting a table by columns so that a group of columns is placed into the new table and the remaining columns are placed in another new table:

- Vertical splitting page 56
- Horizontal splitting
- Adding redundant column
- None of the given option

Vertical splitting involves splitting a table by columns so that a group of columns is placed into the new table and the remaining columns are placed in another new table.

23) OLAP implementations are highly/completely ----- :

- Normalized
- Demoralized page 69

- Predictive
- Additive

OLAP implementations are either highly de-normalized or completely de-normalized,

24) If each cell of Relation R contains a single value (no repeating values) then it is confirmed that :

- Relation R is in 1st Normal Form page 43
- Relation R is in 2nd Normal Form
- Relation R is in 3rd Normal Form
- Relation R is in 3rd Normal Form but not in 2nd Normal Form

Although the table is shown in 1NF i.e. it contains atomic values, there are no repeating values, there is no aggregation, yet it contains redundant data. For example, information

25) Which kind of relationships is captured by Fact less fact table:

- Many- to- Many page 121
- One-to-many
- One-to-one
- None of the given option

are used to capture the relationships between dimensions. Fact less fact table captures the many-to-many relationships between dimensions, but contains no numeric or textual

26) Which of the following is not an example of dimension:

- Product
- Date
- Region
- Sales volume page 78

▪ Dimensions are key business factors for analysis:

- Geographies (city, district, division, province,...)
- Products (item, product category, product department,...)
- Dates (day, week, month, quarter, year,...)

27) Which people criticize Dimensional Modeling (DM) as being a data mart oriented approach?

- Those that consider ER models as Data marts
- Those that consider Business processes as Data marts page 110
- Those that consider Data marts as Data warehouse
- Those that consider dimensional model
- Those that consider dimensional modeling as de-normalization approach

Many people consider business processes as Data marts i.e. in their view organizational or departmental function is referred as the business process. That is why such people criticize DM as being a data mart oriented approach. However, in Kimball's view, it is a

28) In a fully normalized form:

- To many joins are required page 49
- Relationships lose their significance
- No joins are required

- Data integrity becomes an issue

There should be a balance between normalized and de-normalized forms. In a fully normalized form, too many joins are required and in a totally de-normalized form, we

29) Which of the following is an example of Non-Additive Facts:

- Quality sold
- Total sale in Rs.
- Discount in percentage page 119
- Count of orders in a store

- **Non-additive facts:**
 - Averages (average sales price, unit price)
 - Percentages (% discount)

30) Which of the following is not a CUBE operation?

- ANSI SQL page 81
- Roll Up
- Drill Down
- Pivoting

stored in a cube. There are four fundamental cubes operations which are (i) rollup (ii) drill down (iii) slice and dice and (iv) pivoting. In the rollup operation the level of aggregation

31) ----- allows download of "cube" structures to a desktop platform without the need for shared or cube server:

- MPLAP
- ROLAP
- DOLAP page 78
- HOLAP

DOLAP allows download of "cube" structures to a desktop platform without the need for shared relational or cube server. This model facilitates a mobile computing paradigm.

32) ROLAP provides access to information via a relational database using:

- ANSI standard SQL page 78
- Proprietary file format
- Comma Separated Values
- All of the given option

ROLAP or a Relational OLAP provides access to information via a relational database using ANSI standard SQL.

33) ----- is usually deployed when expression can be used to group data together in such a way that access can be targeted to a small set of partitions:

- Expression elimination
- Expression partitioning page 67

- Expression indexing
- None of the given option

Expression partitioning is usually deployed when expressions can be used to group data together in such a way that access can be targeted to a small set of partitions for a significant portion of the DW workload. For example, partitioning a table by expression

34) Taken jointly, the extract programs or naturally evolving systems formed a spider web, also known as

- Distributed Systems Architecture
- Legacy System Architecture page 14
- Online System Architecture
- Intranet System Architecture

Taken jointly, the extract programs or naturally evolving systems formed a spider web, also called "legacy systems" architecture.

35) The data has to be checked, cleaned and transformed into a ----- format to allow easy and fast access

- Unified page 20
- Predicated
- Qualified
- Proactive

data in different formats and focus on different aspects of the subject. The data has to be checked, cleansed and transformed into a unified format to allow easy and fast access.

36) Suppose in a system A, the values of "PhoneNo" attribute were stored in "countrycode-phone-extension" format, however after transformation into data warehouse the separate columns were used for "countrycode", "phone" and "extension". The above scenario is an example of :

- One-to-one scalar transformation
- One-to-many element transformation page 144+conceptual
- Many-to-one element transformation
- Many-to-many element transformation

One-to-many element transformations

- 4 x 20 address field → House/Flat, Road/Street, Area/Sector, City.

element from the source system results in several columns in the DW. Consider the 6x30 address field (6 lines of 30 characters each), the requirement is to parse it into street address lines 1 and 2, city, state and zip code by applying a parsing algorithm.

37) In decision support system ease of use is achieved by:

- Normalization
- Denormalization page no 49
- Drill up
- Drill down

Normalization is a rule of thumb in DBMS, but in DSS ease of use is achieved by way of denormalization.

38) Which of the following is one of the methods to simplify an ER model?

- Normalization
- Denormalization page no 103
- HOLAP
- Hybrid schema

13.5 How to simplify an ER data model?

- Two general methods:
 - De-Normalization
 - Dimensional Modeling (DM)

39) In ETL process data transformation includes -----

- Data cleansing page 129
- Data aggregation
- Behavior checking
- Pattern recognition

the data into the DWH. There is a significant part after data transformation that involves data cleansing i.e. there is a C also here. However, ETL is a pretty standard definition,

40) Non-uniform use of abbreviations, units, and values refers to:

- Syntactically dirty data page 160
- Semantically dirty data
- Coverage anomaly
- Extraction issue

19.3 Syntactically Dirty Data

Lexical errors: For example, assume the data to be stored in table form with each row representing a tuple and each column an attribute. If we expect the table to have five columns because each tuple has five attributes but some or all of the rows contain only four columns then the actual structure of the data does not conform to the specified format.

Irregularities are concerned with the non-uniform use of values, units and abbreviations.

41) Suppose the size of the attribute "Computerized National Card (CNIC) no. is changed in NADRA database. This transformation refers to:

- Format revision page 153
- Field splitting
- Field decoding
- Calculation of derived value

Format Revisions: You will come across these quite often. These revisions include changes to the data types and lengths of individual fields 12 digit new NID. In your

42) The divide and conquer cube partitioning approach helps alleviate the ----- limitations of MOLAP implementation:

- Flexibility
- Maintainability
- Security
- Scalability

page 85

- The divide&conquer cube partitioning approach helps alleviate the scalability limitations of MOLAP implementation.

43) identify the TRUE statement:

- DM is inherently dimensional in nature
- DM comprises of a single central fact table
- DM comprises of a set of dimensional tables
- All of the given option

Page 103

- Inherently dimensional in nature, with a single central fact table and a set of smaller dimensional tables.

44) ----- can be used when some columns are rarely accessed rather than other columns or when the table has wide rows or header or both:

- Horizontal splitting
- Pre-joining
- Vertical splitting
- Derived attributes

page 56

Vertical splitting can be used when some columns are rarely accessed rather than other columns or when the table has wide rows or header or both. Thus the infrequently

45) Which of the following is an example of derived attributes?

- Age
- Size
- Color
- Length

page 61

Age is also a derived attribute, calculated as Current_Date – DoB (calculated periodically).

46) The online high performance transaction processing was evolved in -----:

- 1980
- 1975
- 1977
- 1965

page 12

1975

Online high performance transaction processing

47) Cube is a logical entity containing values of a certain aggregation level at an intersection of a combination of ----- :

- Facts
- Dimension

page 88

- Summary tables
- Primary and foreign key

▪ **Cube is a logical entity containing values of a certain fact at a certain aggregation level at an intersection of a combination of dimensions.**

48) Which of the following is TRUE regarding Entity relationship modeling?

- It does not really model business, but models the micro relationships among data elements.
- ER modeling does not have "business rules," it has "data rules"
- ER modeling helps retrieval of individual records having certain critical identifiers.
- All of the given option page 102

1- ER models are constituted to (a) remove redundancy from the data model, (b) facilitate retrieval of individual records having certain critical identifiers, and (c) therefore, optimize On-line Transaction Processing (OLTP) performance.

2- ER modeling does not really model a business; rather, it models the micro relationships among data elements. ER modeling does not have "business rules," it has "data rules."

49) ----- facilitates a mobile computing paradigm:

- DOLAP page 78
- HOLAP
- ROLAP
- MOLAP

DOLAP allows download of "cube" structures to a desktop platform without the need for shared relational or cube server. This model facilitates a mobile computing paradigm.

50) The main reason(s) for the increase in cube size may be:

- Increase in the number of dimensions
- Increase in the cardinality of the dimensions
- Increase in the amount of detail data
- All of the given options page 87

size increases. There may be many reasons for the increase in cube size, such as increase in the number of dimensions, or increase in the cardinality of the dimensions, or increase in the amount of detail data or a combination of some or all these aspects. Thus there is an

51) Suppose the amount of data recorded in an organization is doubled in year. This increase in -----

- Linear
- Quadratic
- Exponential page 15
- Logarithmic

contents carried. On the practical side, it has been noted that the amount of data recorded in an organization doubles every year and this is an exponential increase.

52) The data in the data warehouse is ----- :

- Volatile

- Non-volatile page 69

- Static
- Non-structured

A data warehouse is a "subject-oriented, integrated, time varying, non-volatile collection of data that is used primarily in organizational decision making. Typically, the data

53) ----- models the macro relationships among data elements with an overall deterministic strategy:

- Dimensional model page 102

- Entity relationship model
- Object oriented model
- Structured model

DM
Constituted to optimize DSS query performance.
Models the macro relationships among data elements with an overall deterministic strategy.

54) De- normalization affects:

- Database size and query performance page 52
- Database usability and query reliability
- Database availability and query success
- None of the given options

De-normalization will ultimately affect the database size and query performance.

55) ----- technique requires a separate column to specify the time and date when the last modification was occurred:

- Checkmarks
- Timestamps page 150
- Just-in-Time

Real Time extraction

Timestamps

The tables in some operational systems have timestamp columns. The timestamp specifies the time and date that a given row was last modified. If the tables in an operational system

56) Which of the de-normalization technique squeezes master table into detail?

- Pre-joining page 58
- Horizontal splitting
- Vertical splitting
- Adding redundant column

To avoid the run-time join, we use the pre-join technique and “squeeze” the sales master information into the detail table. The obvious drawback is repetition of facts from the

57) De-normalization can help:

- Minimize joins
- Minimize foreign keys
- Resolve aggregates
- All of the given options page 51

De-normalization can help minimize joins and foreign keys and help resolve aggregates.

58) The domain of the “gender” field in some database may be (‘F’, ‘M’) or as (“Female”, “Male”) or even as (1, 0). This is:

- Primary key problem
- Non primary key problem page 163
- Normalization problem
- All of the given option

Non primary key problems...

1. Different encoding in different sources.
2. Multiple ways to represent the same information.
3. Sources might contain invalid data.
4. Two fields with different data but same name.

1. Data may be encoded differently in different sources. The domain of a “gender” field in some database may be {‘F’, ‘M’} or as {“Female”, “Male”} or even as {1, 0}.

59) Increasing level of normalization ----- number of tables:

- Increases page 51
- Decreases
- Does not effect
- None of the given option

The higher the level of normalization, the greater will be the number of tables in the DSS

60) Which of the following is not a Data Quality Validation Technique:

- Referential integrity
- Using Data Quality Rules
- Data Histograming
- Indexes page 189

22.2 Data Quality Validation Techniques

- Referential Integrity (RI).
- Attribute domain.
- Using Data Quality Rules.
- Data Histograming.

61) This technique can be used when column from one table is frequently accessed in a large scale join in conjunction with a column from another table:

- Horizontal splitting
- Pre-joining
- Adding redundant column page 58
- Derived attributes

8.3 Adding Redundant Columns

This technique can be used when a column from one table is frequently accessed in a large scale join in conjunction with a column from another table. To avoid this join, the

62) Data cleansing requires involvement of domain expert because

- Domain expert has deep knowledge of data aggregation
- Change Data captures requires involvement of domain expert
- Domain knowledge is required to correct anomalies page 158
- Domain expert has deep knowledge of data summarization

Usually the process of data cleansing can not be performed without the involvement of a domain expert because the detection and correction of anomalies requires detailed domain knowledge. Data cleansing is therefore described as semi-automatic but it should be as

63) Relational databases allow you to navigate the data in ----- that is appropriate using the primary, foreign key structure within the data model:

- Only One Direction
- Any Direction page 19
- Two Direction
- None of these

warehouse, because relational databases allow you to navigate the data in any direction that is appropriate using the primary, foreign key structure within the data model.

64) History is excellent predictor of the -----:

- Past
- Present
- Future page 15
- History

- History is excellent predictor of the future.

65) De-normalization is the process of selectively transforming normalized relations into un-normalized physical record specifications, with the aim to:

- Well structure the data
- Well model the data
- Reduce query processing time page 50
- None of the given option

information will not be lost. De-normalization is the process of selectively transforming normalized relations into un-normalized physical record specifications, with the aim of reducing query processing time. Another fundamental purpose of denormalization is to

66) ----- gives total view of an organization:

- OLAP
- OLTP
- Data Warehouse page 16
- Database

So why would you want data warehouse in your organization? First of all a data warehouse gives a total view of an organization. If you look at the operational system i.e.

67) Suppose in system A, the possible values of "Gender" attribute were "Male"& "Female", however in data warehouse ,the values stored were "M" for male and "F" for female. This above scenario is an example of :

- One-to-one scalar transformation page 144
- One-to-many element transformation
- Many-to-one element transformation
- Many-to-many element transformation

Simple scalar transformation is a one-to-one mapping from one set of values to another set of values using straightforward rules. For example, if 0/1 corresponds to male/female in one source system, then the gender is stored as male/female in the DW.

68) Enrichment is one of the basic tasks in data ----- :

- Extraction
- Transformation page 138
- Loading
- Summarization

Enrichment is one of the basic tasks of data transformation

69) Which of the following is not a technique of De-normalization?

- Pre-joining
- Splitting tables
- Adding redundant columns
- ER modeling page 52

Now we will discuss **de-normalization techniques** that have been commonly adopted by experienced database designers. These techniques can be classified into four prevalent strategies for denormalization which are:

1. **Collapsing Tables.**
 - Two entities with a One-to-One relationship.
 - Two entities with a Many-to-Many relationship.
2. **Pre-joining.**
3. **Splitting Tables** (Horizontal/Vertical Splitting).
4. **Adding Redundant Columns** (Reference Data).

70) Which of the following is an example of Additive Facts?

- **Sales Amount** page 119
- Average
- Discount
- Ratios

There can be two types of facts i.e. additive and non-additive. **Additive facts are those facts which give the correct result by an addition operation. Examples of such facts could be number of items sold, sales amount etc.** Non-additive facts can also be added, but the

71) Robotic libraries are needed for -----

- Cubes
- Data marts
- **Data warehouse** page 131
- Aggregates

use a robotic tape library and all the associated setup. **Typically you don't need robotic libraries for the cubes and data marts, but you do need them for data warehouse. But**

72) Normally ROLAP is implemented using -----

- **Star schema** page 87
- Hybrid schema
- Pre-defined aggregate
- All of the given options

OLAP. ROLAP tools will query the relational database using SQL generated to conform to a framework using the facts and dimensions paradigm using the star schema.

73) The relation R will be in 2nd Normal Form if

- It is in 1NF and each cell contains single value
- **It is in 1NF and each non key attribute is dependent upon entire primary key** page 44
- It is in 1NF and non key attribute is dependent upon a single column of composite primary key
- It is in 1NF and Primary key is composite

A relational table is in second normal form 2NF if it is in 1NF and every non-key column is fully dependent upon the primary key. That is, every non-key column must be

74) In ----- nested loop join of quadratic time complexity does not hurt the performance

- Typical OLTP environments page 22
- Data warehouse
- DSS
- OLAP

In typical OLTP environments, the size of tables are relatively small, and the rows of interest are also very small, as the queries are confined to specifics. Hence traditional joins such as nested-loop join of quadratic time complexity does not hurt the performance

75) In Extract, Load, Transform(ELT) process, data transformation -----:

- Takes place on the data warehouse server page 147
- Takes place on a separate transformation server
- Depends on the nature of the source database
- Does not take place

ELT: Extract, Load, Transform in which data transformation takes place on the data warehouse server.

76) Node of a B-Tree is stored in memory block and traversing a B-Tree involves ----- page faults:

- $O(n \log n)$
- $O(\log n)$ page 22
- $O(n)$
- $O(n^2)$

Trees becomes very large. Typically the node of a B-Tree is stored in a memory block, and traversing a B-Tree involves $O(\log n)$ page faults. This is highly undesirable, because by default the height of the B-Tree would be very large for very large data bases

77) As dimensions get less detailed (e.g., year vs. day) cubes get -----

- Smaller page 84
- Larger
- Partitioned
- Merged

Storage issue: As dimensions get less detailed (e.g., year vs. day) cubes get much smaller, but storage consequences for building hundreds of cubes can be

78) Which of the following is not a technique of "Changed Data Capture" in currently used Modern Source System?

- Timestamps
- Partitioning
- Triggers
- Dimensional Modeling page 150

Several techniques are used for implementing changed data capture in currently used state of the art source systems:

- Timestamps
- Partitioning
- Triggers

79) The trade-offs of de-normalization is/are:

- Storage
- Performance
- Ease-of-use
- All of the given options page 62

The trade-offs of denormalization are as follows:

- Storage
- Performance
- Ease-of-use
- Maintenance

80) If actual data structure does not conform to documented formats then it is called:

- Syntactically dirty data page 160
- Semantically dirty data
- Coverage anomaly
- Extraction issue

19.3 Syntactically Dirty Data

Lexical errors: For example, assume the data to be stored in table form with each row representing a tuple and each column an attribute. If we expect the table to have five columns because each tuple has five attributes but some or all of the rows contain only four columns then the actual structure of the data does not conform to the specified format.

81) "Header size is reduced, allowing more rows per block, thus reducing I/O". The above statement is TRUE with respect to:

- Vertical splitting page 56
- Horizontal splitting
- Adding redundant column
- None of the given options

Splitting Tables: Vertical Splitting

- Splitting and distributing into separate files with repeating primary key.
- Infrequently accessed columns become extra "baggage" thus degrading performance.
- Very useful for rarely accessed large text columns with large headers.
- Header size is reduced, allowing more rows per block, thus reducing I/O.

82) ----- Breaks a table into multiple tables based upon common column values

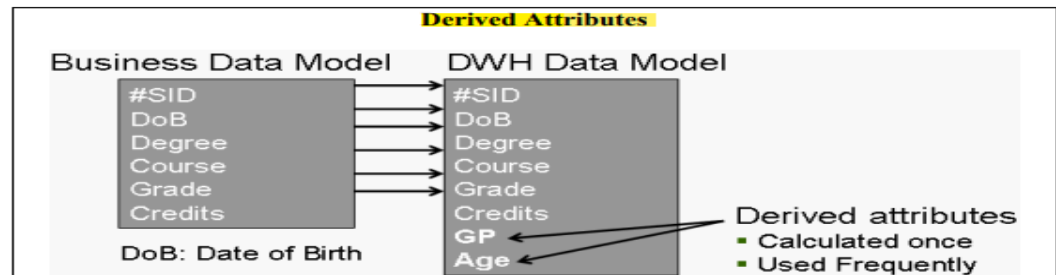
- Horizontal splitting page 54
- Vertical splitting
- Adding redundant column
- None of the given option

Splitting Tables: Horizontal splitting

Breaks a table into multiple tables based upon common column values. Example: Campus

83) Which of the following is NOT an example of derived attribute?

- Age
- CGPA
- Area of rectangle
- Height (Conceptual)



84) Which of the following is NOT an example of derived attribute?

- Age
- CGPA
- Annual Salary
- Email Address (Conceptual)

85) If a table is expected to have six columns but some or all of the records do not have six columns then it is example of.

- Syntactically dirty data page 160
- Semantically dirty data
- Coverage anomaly
- Extraction issue

19.3 Syntactically Dirty Data

Lexical errors: For example, assume the data to be stored in table form with each row representing a tuple and each column an attribute. If we expect the table to have five columns because each tuple has five attributes but some or all of the rows contain only four columns then the actual structure of the data does not conform to the specified format.

86) MDX by Microsoft is an example of -----:

- HOLAP
- DOLAP
- ROLAP

- None of the given options page 79

MOLAP Implementations

- No standard query language for querying MOLAP
 - *No SQL !*
- Vendors provide proprietary languages allowing business users to create queries that involve pivots, drilling down, or rolling up.
 - E.g. MDX of Microsoft

87) The growth of master files and magnetic tapes exploded around the mid- -----

- 1950s
- 1960s page 12
- 1970s
- 1980s

Around the mid-1960s, the growth of master files and magnetic tapes exploded. Soon

88) If one or more records in a relational table do not satisfy one or more integrity constraint , then the data:

- Is syntactically dirty
- Is semantically dirty page 160
- Has Coverage anomaly
- Has extraction issue

Semantically dirty data

Integrity constraint violations describe tuples (or sets of tuples) that do not satisfy one or more of the integrity constraints. Integrity constraints are used to describe our

89) OLAP is:

- Analytical processing page 69
- Transaction processing
- Additive processing
- Active processing

OLAP is Analytical Processing instead of Transaction Processing. It is also NOT a

90) One of the possible issues faced by web scrapping is that:

- Web pages may contain junk data page 141
- Web pages do not contain multiple facts
- Web pages do not contain multiple dimensions
- Web pages does not support transformation

17.3 Web scrapping

- Lot of data in a web page, but is mixed with a lot of "junk".

91) Which of the following is\are example of dimension:

- Product page 79
- Region

- Data
- None of the given

WHERE clause is executed. Some examples of dimension are geography, time, products etc. Note that typically dimensions have hierarchies, for example geography dimension

92) An OLTP system is always good at -----:

- Evolving data page 122
- Keeping static data
- Tracking past data
- Maintaining historic data

discussed before. A good OLTP system is always evolving. Orders are being filled and,

93) In case of multiple sources for the same data element , we need to prioritize the source systems per element based, the process is called:

- Ranking page 143
- Prioritization
- Element selection
- Measurement selection

This means you need to establish ranking in the source system on per element (attribute) basis. Ranking is all about selecting the “right” source system. Rank establishment has to

94) One feature of Change Data Capture (CDC) is that:

- It pre-calculates changed aggregates
- It loads the transformed data in real time
- It only processes the data has been changed
- It can automate the transformation of extracted data page 149

from one or more source databases, into the data warehouse for analysis. Change Data Capture (CDC) quickly identifies and processes only the data that has changed, not entire tables, and makes the change data available for further use.

95) In ----- SQL generation is vastly simplified for front-end tools when the data is highly structure:

- MOLAP
- Star Schema page 107
- Hybrid schema
- Object oriented schema

The ultimate goal of a star schema design is to put into place a physical data model capable of very high performance to support iterative analysis adhering to an OLAP model of data delivery. Moreover, SQL generation is vastly simplified for front-end tools when the data is highly structured in this way.

96) Dirty data means:

- Data cannot be aggregated
- Data contains non-additive facts
- Data does not fulfill dimensional modeling rules

- Data does not conform to proper domain definitions page 158

and you don't even want to consider what work cleaning it up would take. By "dirty," I mean that it does not conform to proper domain definitions or "make sense." The age-old

97) In Context of Change Data Capture (CDC) sometimes a ----- object can be used to store recently modified data:

- Buffer table
- Change table page 149
- Checkmark table
- Change control table

user tables is recorded. For some environments, the change data is then stored in a database object called a *change table*, and the changed data is made available to applications in a controlled way.

98) "Sometimes during data collection complete entities are missed". This statement is an example of :

- Missing tuple page 161
- Missing attribute
- Missing aggregates
- Semantically dirty data

Missing tuples

Result from omissions of complete entities existent in the mini-world that are not represented by tuples in the data collection. Anomalies could further be classified

99) Table collapsing technique is applied in case of:

- One-by-one relation or many-to-many relation page 52
- One-to-many relation
- Many-to-many relation
- None of the given option

1. Collapsing Tables.

- Two entities with a One-to-One relationship.
- Two entities with a Many-to-Many relationship.

100) Which of the following is an example of dimension?

- Product
- Region
- Date
- All of the given option page 78

Dimensions are key business factors for analysis:

- Geographies (city, district, division, province,...)
- Products (item, product category, product department,...)
- Dates (day, week, month, quarter, year,...)

101) Data warehouse stores -----:

- Operational data

- Historical data page 24

- Meta data
- Log files data

customers and old transactions. In the data warehouse we save the historical data.

102) The business process covered by ER diagrams:

- Do not co-exist in time and space page 109
- Co-exist in time and space
- Do not physically exist in real time context
- None of the given options

A typical ER diagram covers the entire spectrum of the business, actually covers every possible business process. However, in reality those multitudes of process do not co-exist in time and space (tables). As a consequence, an ER diagram is overly complex, and is a

103) The main goal of normalization is to eliminate -----:

- Data redundancy page 41
- Data sharing
- Data security
- Data consistency

Normalization is the process of efficiently organizing data in a database by decomposing (splitting) a relational table into smaller tables by projection. There are basically two goals of normalization as follows:

1. Eliminate redundant data (for example, storing the same data in more than one table)

104) Serious ---- involves decomposing and reassembling the data:

- Data cleansing page 168
- Data transformation
- Data loading
- Data extraction

Data cleansing is much more than simply updating a record with good data. Serious data cleansing involves decomposing and reassembling the data. One can break down the

105) In the data warehouse environment the data is -----

- Subject- oriented page 69
- Time- oriented
- Both subject and time oriented
- Neither time-oriented nor subject- oriented

A data warehouse is a "subject-oriented, integrated, time varying, non-volatile collection of data that is used primarily in organizational decision making. Typically, the data

106) For large record spaces and large number of records , the run time of the clustering algorithms:

- Prohibitive page 164
- Static

- Exponential
- Numerical

computational time. The clustering algorithms have high computational complexity. For large record spaces and large number of records, the run time of the clustering algorithms is prohibitive.

107) ----- can result in costly errors, such as , False frequency distributions and incorrect aggregates due to double counting:

- Data duplication page 165
- Data reduction
- Data anomaly
- Data transformation

Data duplication can result in costly errors, such as:

- False frequency distributions.
- Incorrect aggregates due to double counting.

108) The degree to which values are present in the attributes that require them is known as -
-----:

- Completeness page 185
- Uniqueness
- Accessibility
- Consistency

Completeness	The degree to which values are present in the attributes that require them.
--------------	---

109) Time complexity of Key Creation process in basic Sorted Neighborhood (BSN) Method is
-----:

- $O(n \log n)$
- $O(\log n)$
- $O(n)$ page 171
- $O(2n)$

Complexity Analysis of BSN Method

- Time Complexity: $O(n \log n)$
 - $O(n)$ for Key Creation
 - $O(n \log n)$ for Sorting
 - $O(w n)$ for matching, where $w \leq 2 \leq n$
 - Constants vary a lot

110) Which of the following is an example of slowly changing dimensions?

- Inheritance page 124
- Aggregation
- Association
- Asset disposal

"Slowly" Changing Dimensions.

There can be many examples. For a young customer who is single, then after a while the customer gets married. After sometime there are children, in unfortunate cases the marriage breaks so the customer is separated or the husband dies and the customer becomes a widow. This just does not typically happen overnight but takes a while. Another example is inheritance, consider the example of land. Over a period of time the

111) The ----- operator proves useful in more complex metrics applicable to the dimensions and accessibility:

- Max page 188
- Min
- Max and Min
- None of the given

believability and appropriate amount of data. The max operator proves useful in more complex metrics applicable to the dimensions of timeliness and accessibility.

112) In OLAP, the typical write operation is ----- :

- Bulk insertion page 75
- Single insertion
- Sequential insertion
- No insertion

Typical write operation	Update, insert, delete	Bulk insert, almost no deletion
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113) The issue(s) of "Adding redundant column" includes(s):

- Increase in table size
- Maintenance
- Loss of information
- All of the given option page 65

9.3 Other Issues: Adding redundant columns

Other issues include, increase in table size, maintenance and loss of information:

114) ----- is applicable in Profitability analysis:

- OLTP
- Data warehouse page 36,37
- Information System(IS)
- Management Information System(MIS)

5.3 Typical Applications of DWH

There are, and there can be many many applications of a data warehouse. It is not possible to discuss all of them. Some representative applications are listed to be discussed as follows:

- Fraud detection.

- Profitability analysis.
- Direct mail/database marketing.
- Credit risk prediction.
- Customer retention modeling.
- Yield management.
- Inventory management.

115) The hardware (CPU) utilization in data warehouse environment is full or ----- :

- Fixed
- Partial
- Not at all page 24
- Slow

relatively static pattern of utilization. There is an essentially different pattern of hardware utilization in the data warehouse environment i.e. a binary pattern of utilization, either the hardware is utilized fully or not at all. Calculating a mean utilization for a DWH is not a

116) Time variant is a characteristics of data warehouse which means:

- Data loaded in data warehouse will be time stamped page 20
- Data can be loaded in data warehouse anytime
- Data can be loaded in data warehouse only at a particular time
- Data cannot be loaded in data warehouse with respect to time

Time variant: Time variant records are records that are created as of some moment in time. Every record in the data warehouse has some form of time variability associated with it. In an OLTP system, the contents change with time i.e. updated such as bank account balance or mobile phone balance, but in a warehouse as the data is loaded; the moment usually becomes its time stamp.

117) In which class of aggregates AVERAGE function can be placed:

- Algebraic page 120
- Distributed
- Associative
- Holistic

- Algebraic
 - Compute aggregate from constant-sized summary of subgroup
 - Examples: STDDEV, AVERAGE
 - For AVERAGE, summary data for each group is SUM, COUNT

118) Considered the following Employee table and identify the column which causes that the table is not in first normal form(1NF):

Employee(Emp_ID, Emp_Name ,Emp_skills, Emp_Designation)

- Emp_ID
- Emp_Name
- Emp_skills page 43(conceptual)

- Emp_Designation

Although the table is shown in 1NF i.e. it contains atomic values, there are no repeating values, there is no aggregation, yet it contains redundant data. For example, information

119) The application of data and information leads to -----

- Intelligence
- Experience
- Knowledge page 11
- Power

questions such as "who", "what", "where", and "when". Knowledge, on the other hand is an application of information and data, and gives an insight by answering the "how"

120) ----- segregate data into separate partitions so that queries do not need to examine all data in a table when WHERE clause filters specify only a subset of the partitions.

- Pre-joining technique
- Collapsing table technique
- Horizontal splitting technique page 56
- Vertical splitting technique

Goals of Horizontal splitting:

There are typically two specific goals for horizontal partitioning: (1) spread rows in a large table across many HW components (disks, controllers, CPUs, etc.) in the environment to facilitate parallel processing, and (2) segregate data into separate partitions so that queries do not need to examine all data in a table when WHERE clause filters specify only a subset of the partitions. Of course, what we would like in an ideal